DUE2LAB SRL

To the attention of:
INFN – CSN5
Young Researchers Grant Selection Committee

Subject: Letter of support for Dr Simone Manti's PRISM project application to the INFN–CSN5 Young Researchers Grant programme

Due2Lab S.r.l. is pleased to express its support for PRISM ("Precision Room-temperature Instrumentation for high-energy X-ray absorption Spectroscopy Measurements"), a project led by Dr Simone Manti of INFN – Laboratori Nazionali di Frascati.

Due2Lab designs and manufactures CdZnTe (CZT) sensors and detector systems for X-ray and gamma-ray applications, including single-pixel devices, linear arrays and pixel matrices. From this perspective, we recognize a strong technological connection with PRISM, which aims to establish a laboratory fluorescence-XAS platform above 20 keV based on optimized crystal optics and a custom multichannel CZT detector operating at room temperature.

PRISM addresses detector requirements that are especially relevant to the development of advanced CZT instrumentation: high hard-X-ray efficiency, stable spectroscopic response, channel-to-channel uniformity, operation at useful counting rates, and control of charge-collection, dead-time and pile-up effects. The project's plan to characterize the complete detection chain, from the sensor and low-noise front-end electronics to calibration and waveform-based spectral reconstruction, can provide valuable knowledge for future scientific and industrial applications of segmented CZT systems.

We also consider the proposed integration of the detector with a compact laboratory spectrometer to be of particular interest. Demonstrating quantitative fluorescence-XAS on reference and realistic samples would extend the application of room-temperature CZT technology beyond stand-alone radiation detection and towards complete analytical instrumentation for materials characterization.

Due2Lab is pleased to support the scientific and technological objectives of PRISM and is interested in following its progress and exploring possible future technical exchanges concerning CZT sensor architectures, detector characterization and application requirements. Any specific contribution or collaboration would be discussed separately and subject to mutual agreement.

We believe that PRISM can strengthen the link between semiconductor-detector development and practical hard-X-ray spectroscopy, creating valuable opportunities for research, innovation and technology transfer.

Please do not hesitate to contact us for any further information.

Yours sincerely,

Scandiano, 28/07/2026

Nicola Zambelli

General Manager
due2lab s.r.l.

DUE2LAB SRL
Sede legale e Sede Operativa: Via Borsellino 2, 42019 - Scandiano - Reggio Emilia (ITALY)
Tel: +39 0522 1607010 / Fax: +39 0522 1601375, P.IVA, C.F. e Reg.Imp. Parma: 02709820340 - R.E.A. Parma: 260986 - Cap. Soc.: € 60.000 i. v.
due2lab.com – info@due2lab.com

Dear Sir/Madam,

Subject: Letter of support for Dr Simone Manti's PRISM project application to the INFN-CSN5 Young Researchers Grant programme

It is with great enthusiasm that Rigaku extends its full support for PRISM ("Precision Room-temperature Instrumentation for high-energy x-ray absorption Spectroscopy Measurements"), an innovative project led by Dr Simone Manti of INFN – Sezione di Frascati.

We have carefully considered the project's objectives, methodology, and expected outcomes, and we firmly believe that this initiative holds significant potential for advancing laboratory X-ray absorption spectroscopy. In particular, extending fluorescence-mode XAS above 20 keV would address an important technological gap and enable measurements on thick, dilute, heterogeneous, and in-situ or operando samples that are difficult to investigate in transmission geometry.

From an industrial perspective, the compact and accessible instrumentation envisioned by PRISM could open new opportunities for laboratory analytical systems, routine materials characterization, and long-duration studies of catalytic, electrochemical, and energy-related processes. The project therefore offers a valuable connection between advanced detector research and future practical X-ray instrumentation.

We are convinced that the successful completion of PRISM will contribute significantly to both scientific progress and technological innovation in hard-X-ray spectroscopy. Rigaku is pleased to support this proposal and is interested in following its development and exploring possible future technical exchanges and collaborations.

Please do not hesitate to contact us for any further information or to discuss how Rigaku may contribute to bringing this project to fruition.

Yours sincerely,

Dr. Peter Oberta, Ph.D.
Managing Director
Rigaku Innovative Technologies Europe s.r.o.

Quantum Design
EUROPE

Quantum Design s.r.l.
Società Unipersonale Soggetta a Direzione e Coordinamento
Sede legale e operativa: Via di Grotta Perfetta, 643 - 00142 Roma
+39 06 5004204 r.a. +39 06 5010389 SDI: W7YVJK9
www.qd-europe.com italy@qd-europe.com PEC: qd@pecimprese.it
P.IVA/C.F. 10563320158 R.E.A. RM-818890 Cap. Soc. € 96.900,00 i.v.

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30 luglio 2026

To the attention of the INFN-CSN5
Young Researchers Grant Selection Committee
Dear Members of the Selection Committee,

Subject: Letter of support for Dr Simone Manti's PRISM project application to the INFN-CSN5 Young Researchers Grant programme

It is with great enthusiasm that Quantum Design expresses its support for PRISM ("Precision Room-temperature Instrumentation for high-energy x-ray absorption Spectroscopy Measurements"), an innovative project led by Dr Simone Manti of INFN – Sezione di Frascati.

We have carefully considered the project's objectives, methodology, and expected outcomes, and we believe that PRISM has significant potential to advance laboratory X-ray absorption spectroscopy. The project aims to develop and validate a fluorescence-mode XAS platform for measurements above 20 keV, combining optimized crystal optics with efficient multichannel CZT detection at room temperature.

From an industrial perspective, the compact and accessible approach pursued by PRISM could create new opportunities for advanced materials characterization, quality control, and the study of catalytic, electrochemical, and energy-related processes. The project also offers a valuable bridge between detector and instrumentation research and future practical applications in hard-X-ray spectroscopy.

We are convinced that the successful completion of PRISM will contribute to both scientific progress and technological innovation. Quantum Design is pleased to support this proposal and is interested in following its development and exploring possible future technical exchanges and collaborations.

Please do not hesitate to contact us for any further information.

Yours sincerely,

Signature:

By: Diego Vitaglione

Title: Amministratore Delegato

Quantum Design S.r.l.

Date: Roma 30/07/2026

Quantum Design S.r.l.
Via di Grotta Perfetta 643
00142 Roma
Partita Iva 10563320158